Load Testing as a Service

Performance Testing as a Service

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1. Executive Summary

Yntraa Cloud Load Testing enables you to generate high-scale load and simulate traffic for your applications, regardless of where they're hosted. It supports running **Apache JMeter-based tests-based** tests. It also enables generating loads from multiple regions and enables you to test private application endpoints. It provides detailed metrics and insights into the performance of your application underload, helping you identify bottlenecks and optimize performance.

From your workspaces or logged into Yntraa Cloud console to run end-to-end tests with high parallelization. It supports running tests in parallel across multiple browsers and devices, enabling you to validate the functionality and performance of your applications at scale. It also provides detailed test results and insights, helping you identify issues and optimize your tests.

Benefits of Yntraa Cloud Service:

- **Managed Testing Infrastructure:**
  Yntraa manages the complete load testing environment, including load generator instances, orchestration components, and supporting infrastructure, eliminating the need for customers to provision or maintain testing tools and servers.
- **Elastic Load Generation:**
  Customers can easily scale load testing capacity by adding additional load generator instances to simulate thousands of concurrent users or high transaction volumes.
- **High Availability Testing Platform:**
  The distributed architecture ensures reliable execution of load tests through coordinated load generator nodes and centralized orchestration, enabling consistent and uninterrupted testing workflows.
- **Secure Test Execution:**
  Yntraa ensures secure connectivity between load generator infrastructure and the target environment using controlled network access, secure authentication mechanisms, and adherence to platform security standards.
- **Geographic Load Simulation:**
  Load generation can be configured from different infrastructure zones within Yntraa Cloud to simulate real-world user traffic patterns and geographic access scenarios.

This document serves as a reference and specification guide for evaluating **Load Testing as a Service** offering. It outlines service features, scope of service, operational benefits, and guidance to determine if selecting Load Testing is the right Performance Testing as a service to meet your business requirements.

2. Why Choose Load Testing as a Service

Organizations today operate applications that must remain responsive and reliable under unpredictable and rapidly increasing user demand. Traditional performance testing approaches struggle due to high infrastructure costs, specialized skill requirements, tooling complexity, and limited scalability. Yntraa Cloud's Load Testing as a Service (LTaaS) addresses these limitations by providing a fully managed, subscription-based performance testing platform combined with expert operational support.

LTaaS enables customers to validate real-world system behavior under concurrent user load, stress conditions, and peak traffic events without the overhead of purchasing, configuring, and maintaining dedicated test environments.

1. **Zero Infrastructure Management**

All compute resources, load generator agents, configurations, and runtime environments are provisioned, maintained, and monitored by Yntraa. Customers do not have to invest in hardware, licenses, or support systems.

2. **On-Demand Scalability**

Load capacity can be scaled instantly by adding additional load generator instances. This allows realistic simulation of thousands to hundreds-of-thousands of virtual users during:

- Peak business periods
- UAT cycles
- Go-live events
- Marketing campaigns

3. **Access to Performance Engineering Expertise**

Load testing requires specialized skills to design scenarios, parameterize scripts, correlate responses, measure KPIs, and interpret results. Yntraa provides dedicated engineering support to ensure accurate test design and actionable findings.

4. **Faster Release and UAT Cycles**

Teams can validate application performance during each release cycle without waiting for capacity or specialist availability. This reduces go-live delays and minimizes the risk of performance regressions.

5. **Cost-Efficient Delivery Model**

Customers only subscribe to the required load generator instances. There is no need to hire performance engineers, procure servers, or maintain tools for occasional testing needs.

6. **Detailed Reporting and Analysis**

Post-execution reports include Response time (p50/p95/p99), Throughput, Error percentage, Resource utilization, Bottleneck identification, Recommendations for optimization.
This accelerates troubleshooting and remediation.

7. **Continuous Application Performance Assurance**

Performance is not static. As application code, configurations, and dependencies evolve, continuous load testing ensures sustained user experience over time.

8. **No Procurement Complexity**

The subscription model eliminates repeated purchase approvals for each test run. This simplifies operations and increases testing frequency.

9. **Improved Production Stability**

Pre-production performance validation significantly reduces the risk of:

- Outages on go-live day
- Downtime during high-traffic events
- Customer impact and reputational damage

3. When To Select MySQL

Yntraa Load Testing as a Service is recommended when organizations need to validate the performance, scalability, and resilience of their applications under realistic or extreme user load conditions. LTaaS is ideal in any scenario where application behavior, response times, throughput, and stability must be assessed before production deployment or planned business events.

1. **Pre-Go-Live Readiness**

Before launching a new application, module, feature, or architectural change, LTaaS ensures the environment can handle anticipated user traffic without performance degradation.

2. **User Acceptance Testing (UAT) with Performance Requirements**

During UAT phases, performance baselines must be validated alongside functional testing to ensure production readiness.

3. **Scaling Applications for Peak Traffic Events**

Applications expecting increased traffic due to: Seasonal demand, Promotional campaigns, Government deadlines, Public registration periods. should undergo load testing to avoid outages and slowdowns.

4. **Post-Migration Validation**

After migrating applications to New cloud environments or New infrastructure or New database versions or Refactored microservices or Container/Kubernetes platforms, Then Load testing verifies that performance remains consistent or improved.

5. **Performance Regression Testing**

New releases, updates, or code changes may introduce latency or resource bottlenecks. LTaaS helps detect performance regressions early.

6. **API, Microservices, and Gateway Stress Validation**

Distributed architectures must be tested for: Concurrent request handling, Circuit breaker thresholds, Rate limiting behaviour, Latency under high throughput. Then LTaaS simulates these patterns effectively.

7. **Capacity Planning and Infrastructure Sizing**

Organizations can determine: Optimal VM sizes, Required node counts, Auto-scaling thresholds, Right-sizing opportunities. Then this reduces cloud and hardware costs.

8. **Disaster Recovery and Failover Testing**

Load testing helps validate: Multi-AZ performance, Failover response times, Database replication lag behavior, End-user experience during DR events.

9. **Third-Party Dependency Validation**

When applications rely on: Payment gateways, KYC systems, SMS/Email services, Identity providers. Then LTaaS identifies how these external dependencies behave under stress.

10. **Application Modernization Initiatives**

When shifting from: Monoliths to microservices, Legacy to containerized workloads, On-prem to cloud-native stacks. Then Load testing ensures architectural improvements translate to performance gains.

11. **Compliance and Governance Requirements**

Regulated sectors such as BFSI, Telecom, and Government often mandate periodic performance testing for audit and SLA validation.

12. **Production Incident Post-Mortem**

After an outage, LTaaS can: Reproduce load conditions, Identify performance bottlenecks, Validate fixes before re-deploying

4. Service Overview

Yntraa Cloud Load Testing as a Service (LTaaS) provides a fully managed and scalable platform for simulating real-world user load on applications, APIs, databases, and microservices. The service eliminates the need for customers to procure infrastructure, install testing tools, manage distributed load generators, or hire performance engineering specialists.

The service combines **cloud-native load generation infrastructure**, **industry-standard testing tools**, and **expert-driven performance engineering processes** to deliver end-to-end load testing—from planning and test design to execution, monitoring, reporting, and optimization recommendations.

The service is ideal for government workloads, public portals, financial systems, telecom platforms, API-driven services, and enterprise applications where reliable performance and scalability are essential. With a consumption-based and instance-based model, customers can choose the exact scale required, adding load generator instances as needed to simulate thousands of concurrent users.

Yntraa LTaaS integrates seamlessly into UAT cycles, DevOps pipelines, pre-production validation, and post-migration testing, providing a complete performance assurance capability without operational overhead or licensing complexity.

4.1. Scope of Service for Load Testing

The load testing process under Yntraa Cloud Load Testing as a Service (LTaaS) follows a structured, end-to-end methodology that ensures accurate simulation of user behavior, measurable performance insights, and actionable recommendations. Each stage is designed to minimize customer effort while ensuring test quality, repeatability, and alignment with real-world traffic patterns.

- **Customer Inputs / Responsibilities**
  - Provide Target Nodes / Endpoints
  - Define Performance Goals
- **Yntraa Cloud Tasks (Service)**
  - **Requirement Gathering & Test Planning:**
    The engagement begins with collecting functional, technical, and business inputs about the application under test.
    This includes:
    - Expected load and peak concurrency
    - User journey flows and critical transactions
    - API endpoints, request payloads, authentication requirements
    - Target environment details (UAT, Pre-Prod, DR, etc.)
    - Performance KPIs (response time, error rate, throughput)
    - Success criteria and acceptance benchmarks

    A formal **Test Plan** is prepared and reviewed with the customer before proceeding.

  - **Test Script Design,Creation & Validation**
    Based on the identified user flows, performance engineers:
    - Create scripts simulating real user transactions
    - Parameterize dynamic data
    - Configure logic controllers, think-time, correlation, assertions
    - Define scenario pacing, ramp-up, ramp-down, and test duration
    - Validate scripts for accuracy and consistency

    Scripts undergo a **dry run** to ensure functional correctness.

  - **Environment Preparation & Load Generator Setup:**
    Yntraa Cloud provisions the required load generator VM instances as per sizing guidelines.
    This includes:
    - Installing and configuring load testing tools
    - Ensuring connectivity to the application under test
    - Configuring distributed mode (master + load generators)
    - Pre-validating system metrics collection (CPU, RAM, network, DB metrics)

    The environment is stabilized before execution.

- **Test Execution & Live Monitoring:**
  Performance engineers execute the test as per the approved schedule. Activities include:
  - Running load tests
  - Monitoring system behavior in real time
  - Tracking bottlenecks such as CPU spikes, memory leaks, DB locks, slow queries, or API timeouts
  - Capturing logs, errors, metrics, and performance indicators

- **Data Collection & Result Consolidation**
  After the test completes, results from multiple sources are consolidated into a single view:
  - Response time statistics (P50/P90/P95/P99)
  - Error codes and failure patterns
  - Throughput and concurrency
  - Resource utilization (application servers, DBs, caches, APIs)
  - Network latency and dependency performance
  - System logs and alerts triggered during the test

  All data is normalized for report generation.

- **Performance Analysis & Identification of Bottlenecks**
  Yntraa performance engineers analyze:
  - Slow transactions
  - API latency spikes
  - Database locks, buffer issues, slow queries
  - Infrastructure saturation points
  - Resource constraints
  - Application-level errors
  - Scaling inefficiencies or misconfigurations

  Root causes are documented with evidence.

- **Recommendations & Optimization Guidance**
  Based on findings, actionable recommendations are provided, such as:
  - Code optimization
  - DB query/indexing improvements
  - VM size, autoscaling, or capacity adjustments
  - Caching and connection pool tuning
  - API gateway and load balancer configuration improvements
  - Network and dependency-level enhancements

  These recommendations help ensure production readiness.

- **Final Reporting & Review Discussion**
  A comprehensive Performance Test Report is shared, covering:
  - Summary of test objectives and scenarios
  - Key observations and bottlenecks

- Detailed graphs and metrics
- Success vs. defined acceptance criteria
- Optimization recommendations
- Next steps and retest requirements

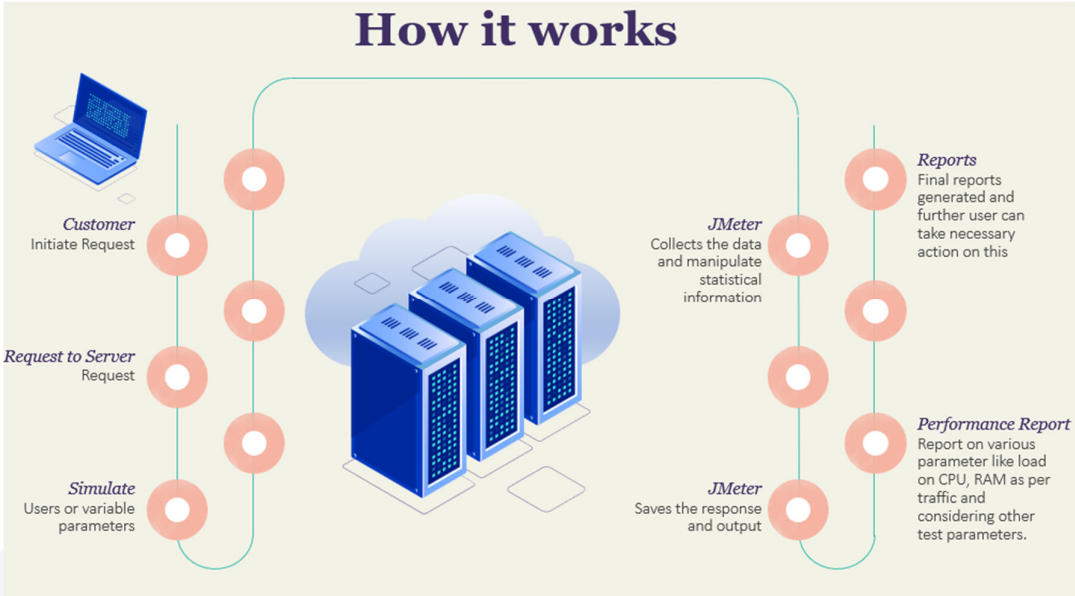
A review meeting is conducted with customer stakeholders.

- **Customer Receives**
  - Performance Reports / Insights / Recommendations

Note: *For escalation models and detailed managed services framework, please refer to the* ***Yntraa Cloud Assure PSS****.*

4.2. How it Works

Yntraa Load Testing as a Service (LTaaS) operates on a cloud-native, scalable, and fully managed architecture that enables the generation of large-scale user traffic without requiring customers to maintain tools, infrastructure, or performance engineering expertise. The service uses distributed load generator instances, orchestrated test execution, and real-time monitoring to simulate real-world application usage scenarios.



The Load Testing test engines abstract the required infrastructure for running a high scale load test. Each test engine instance runs your test script to simulate many virtual users simultaneously accessing your application endpoints. When you create a load test based on a URL (quick test), Load Testing automatically generates a test script for you. To scale out the load test, you can configure the number of test engines. You can host the application under load anywhere: in Yntraa cloud, on-premises, or in other clouds.
To run a load test for services that have no public endpoint, deploy Yntraa cloud Load Testing in a virtual network. During the load test, Load Testing collects the following

resource metrics and displays them in a dashboard like client-side metrics and Server-side metrics. Load tests are automatically stopped if the application endpoints or cloud components start throttling requests.

5. Service Details

5.1. Service Features

Yntraa Load Testing as a Service includes a comprehensive set of capabilities designed to simulate realistic workloads, analyze performance behavior, and deliver actionable insights. The core features include:

1. **Fully Managed Load Testing Platform**

End-to-end setup and operation of all load generator instances, orchestration components, and test execution environments.

2. **Distributed Load Generation**

Horizontally scalable load generator nodes capable of simulating thousands of concurrent users across multiple scenarios.

3. **Script Development & Scenario Modeling**

Creation of test scripts based on critical transactions, user flows, and API calls using industry-standard tools.

4. **Secure Connectivity to Target Environments**

Controlled traffic routing to customer UAT, Pre-Prod, DR, or staging environments through secure channels.

5. **Real-Time Monitoring & Observability**

Live dashboards for monitoring:

- Response times
- Throughput
- Error rates
- CPU/RAM/network on app & DB servers
- Load generator health

6. **Root Cause & Bottleneck Analysis**

Deep analysis of system behavior across application, database, network, and infrastructure layers.

7. **Comprehensive Reporting**

Detailed performance test reports including:

- KPIs

- Graphs & trends
- Bottlenecks & saturation points
- Optimization recommendations

9. Flexible Instance-Based Consumption Model

Customers can scale their load capacity by adding more load generator instances.

5.2. Service Benefits

Industry	Target Segment	Use Cases	Key Benefits
Government & Public Sector	Citizen service platforms, National portals, State departments, NIC	Exam registrations, certificate issuance portals, application submission systems, deadline-based traffic spikes	Prevents portal crashes during peak load, validates DR readiness, ensures SLA compliance, improves citizen experience
Banking, Financial Services, Insurance (BFSI)	Banks, insurance portals, payment processors, fintech platforms	High-volume transactions, UPI/API bursts, loan applications, KYC flows	Ensures transaction stability, validates API concurrency, reduces failures during peak hours, supports regulatory compliance
Telecom & Digital Services	Telcos, DTH providers, OTT platforms	Mass onboarding events, SIM activation portals, plan migration systems	Handles subscriber spikes, reduces downtime during mass events, validates backend APIs, improves customer onboarding
E-Commerce & Retail	Online retailers, marketplaces, D2C brands	Flash sales, festive season traffic, checkout flows, search & catalog load	Ensures high availability during sales, prevents checkout failures, improves conversion rates, handles peak bursts
Education & Examination Systems	Universities, exam boards, online learning portals	Admission portals, exam results, online testing systems	Ensures portal uptime during result days, prevents slowdowns, validates high concurrency, supports student experience
Healthcare & Pharma	Hospitals, digital health apps, insurance TPAs	Appointment systems, patient onboarding, claims processing	Ensures application reliability during emergencies, validates high traffic for critical services, improves

			throughput
IT/ITeS & SaaS Providers	Product companies, API providers, B2B SaaS	API stress testing, multi-tenant architectures, onboarding bursts	Ensures platform scalability, validates microservices, supports multi-tenant workloads, prevents regression failures
Travel, Transport & Hospitality	Airports, hotels, booking portals	Ticket booking, reservation systems, check-in systems	Maintains API and portal performance during travel peaks, prevents booking failures, improves customer satisfaction
Energy & Utilities	Electricity boards, gas/water companies, billing portals	Billing cycles, payment bursts, consumer login spikes	Handles surge loads, prevents outages during billing periods, ensures regulatory SLA adherence
Enterprise Applications (General)	Internal business portals, HRMS, ERP, CRM, custom apps	Release validation, post-migration testing, DR drills	Reduces deployment risk, validates infra changes, ensures stable user experience, supports digital transformation

5.3. Customer Responsibility for MDBaaS

Category	Customer Responsibilities	Yntraa Responsibilities
Environment Access	Provide access to the target testing environment (UAT, Pre-Production, DR, or staging). Ensure required network connectivity and firewall permissions are enabled for load generator instances.	Configure load generator instances and ensure secure connectivity to the customer environment for executing tests.
Application Information	Provide application architecture details, user flows, API documentation, endpoints, authentication methods, and any relevant technical information required to design test	Review application details and design appropriate load testing strategies, scenarios, and execution plans.

	scenarios.	
Test Data & Credentials	Provide valid test user accounts, authentication credentials, and sample data required to simulate real-world user behavior during testing.	Use the provided test data to develop scripts and execute test scenarios accurately.
Test Scenario Inputs	Identify critical user journeys, business transactions, and peak usage expectations to be simulated during load testing.	Convert identified business flows into performance test scripts and realistic load scenarios.
Environment Stability	Ensure the target environment remains stable and available during the scheduled test window and that no conflicting activities are performed during testing.	Coordinate test schedules and execute tests in accordance with the agreed plan while monitoring system behaviour.
Stakeholder Availability	Ensure application owners, developers, and SMEs are available during testing windows for troubleshooting and clarifications if required.	Monitor tests in real time, capture system metrics, and coordinate with customer teams in case of anomalies or failures.
Infrastructure Monitoring	Provide access to relevant application and infrastructure logs where deeper root cause analysis may be required.	Capture and analyze load test results, correlate application performance with infrastructure metrics, and identify bottlenecks.
Remediation & Optimization	Implement recommended improvements such as code changes, configuration tuning, infrastructure scaling, or database optimizations based on test findings.	Provide detailed reports, performance insights, and optimization recommendations based on test results.
Retesting	Request retesting after	Execute retesting (if

Requests	implementing recommended changes or optimizations.	applicable) to validate improvements and confirm system performance stability.
Compliance & Approval	Ensure appropriate approvals are obtained before performing load tests on critical systems or environments.	Ensure load testing is executed safely without impacting production systems unless explicitly approved by the customer.

6. Service Plan

Yntraa Cloud offers Load Testing as a Service under its Performance Testing as a Service (PTaaS) delivered through an **instance-based subscription model**, where customers consume dedicated load generator instances provisioned on Yntraa Cloud. Each instance provides a defined load generation capacity, fully managed tooling, and end-to-end support for load testing activities.

6.1. Service Components

LTaaS includes the following components for what you are charged and getting:

- **Infrastructure**
  - Provisioning of load generator instances
  - High-availability backend components (controller/orchestrator)
  - Tool installation and configuration
  - Network connectivity to application under test
- **Managed Activities**
  - Test planning and scenario design
  - Script creation & parameterization
  - Load test execution
  - Real-time monitoring
  - Consolidation of logs & metrics
  - Detailed reporting
  - Optimization recommendations
- **Support**
  - Engineering assistance
  - Troubleshooting during execution
  - Support for retesting
  - Best practices guidance

6.2. Service Model

- **Per Load Generator Instance (per month)**

  Each instance includes:

- Pre-configured load testing tools
- Defined VU generation capacity
- Unlimited test executions (as per customer plan)
- Fully managed infrastructure & orchestration
- Support from performance engineering team
- Reporting and analysis

Note: *Customers can add or remove instances based on required concurrency and testing scale.*

- **Instance Specifications**
  - Specs: 4 vCPU / 32GB RAM
  - Can simulate: ~500–1,000 concurrent virtual users (HTTP/Web).
  - Runs: One JMeter agent JVM.

  Note: *Actual virtual user capacity may vary depending on test script complexity, response time, think time, payload size, and protocol behavior.*
- **Horizontal Scale-Out Model**

  Concurrency is scaled by adding more instances:

  - 1 instance ≈ up to 1,000 VUs
  - 2 instances ≈ up to 2,000 VUs
  - 5 instances ≈ up to 5,000 VUs
  - 10+ instances for very large tests

  Instances operate in a distributed load generation cluster controlled by a central orchestrator.

Note:
The **total cost of ownership (TCO)** for MySQL DBaaS is calculated as:

Total DBaaS Cost = Database SKU (License + Managed Services) + Infrastructure (vCPU/RAM/Storage/Network/Security/OS)

7. Licensing & Sizing Guidelines

Yntraa Load Testing as a Service is delivered through an **instance-based subscription model**, where each load generator instance contributes a defined load generation capacity. Proper sizing ensures accurate simulation of concurrent users, reliable test execution, and optimal infrastructure utilization.

Sizing depends on multiple factors including the number of concurrent users, complexity of test scripts, response times of the application under test, and the nature of the workload being simulated.

The service follows a **per-instance licensing model**.

Parameter	Description
Unit of Measure	Per Load Generator Instance
Billing Frequency	Monthly subscription
Instance Ownership	Dedicated to customer during subscription period
Test Execution	Unlimited test runs within subscription
Scaling Model	Horizontal scale-out by adding instances

Each instance includes the infrastructure, testing tool configuration, and managed services required to execute load tests.

Sizing Guidance

To determine the number of required instances, the following calculation approach can be used.

Step 1 — Identify Expected Peak Concurrency

Determine the maximum number of simultaneous users expected.

Example:
Peak expected traffic = **5,000 concurrent users**

Step 2 — Determine Per Instance Capacity

Assume average capacity = **1,000 VUs per instance**

Step 3 — Calculate Required Instances

Required Instances =
Peak Concurrent Users ÷ Capacity per Instance

Example:
5,000 users ÷ 1,000 users per instance = **5 instances**

Note: *Actual virtual user capacity may vary depending on test script complexity, response time, think time, payload size, and protocol behavior.*

8. Sizing Best Practices

To ensure accurate load simulation and reliable test execution, the following best practices are recommended:

- Start with baseline load tests before running peak simulations
- Avoid overloading a single instance beyond recommended VU limits
- Gradually increase load in stages to observe performance behavior
- Use distributed load generators for large concurrency tests
- Validate scripts with smaller loads before full-scale testing

Example Sizing Scenarios

Application Type	Expected Concurrent Users	Recommended Instances
Government Registration Portal	3,000 users	3 instances
E-commerce Flash Sale	10,000 users	8–10 instances
Banking API Gateway	5,000 API requests	4–5 instances
Enterprise Internal Portal	1,000 users	1 instance

Factors Affecting Instance Sizing

The number of required load generator instances may vary based on the following workload characteristics:

Factor	Impact on Sizing
Script complexity	More complex logic reduces VU capacity per instance
Request payload size	Larger payloads increase CPU/network usage
Response time	Slow responses increase active thread count
Think time	Higher think time increases VU efficiency
Protocol type	WebSocket, streaming, or encrypted requests consume more resources
Assertion and validation logic	Additional validations increase processing overhead

9. Escalation

Incident handling and escalation for this service follow the standard **Yntraa Cloud Assure** escalation framework. Infrastructure, migration tools, third-party licenses, and value-added services are not part of the base managed service and are chargeable separately. For detailed escalation models, SLAs, and inclusions/exclusions, please refer to Cloud Assure.

10. Explore More with Yntraa Cloud

Yntraa Cloud is India's leading hyperscale cloud platform, delivering secure, scalable, and high-performance infrastructure and platform services. With a wide portfolio ranging from Compute, Storage, Networking, AI/ML, Security, Database-as-a-Service (DBaaS), and many more. Yntraa Cloud empowers enterprises to accelerate digital transformation while ensuring cost efficiency and compliance.

Load Testing as a Service is just one part of our broader cloud ecosystem designed to simplify complex IT needs. Whether you are modernizing applications, scaling workloads, or building next-gen digital platforms, Yntraa Cloud offers the flexibility and reliability you need.

👉 To learn more about Yntraa Cloud and explore our full portfolio of services, visit: **www.yntraacloud.ai**

Revision History

Next Review Date: 01-Apr-26

Version #	Details	Effective Date	Last Review Date	Created By	Approved by
1.0	Initial Document	01-Apr-25	NA	Lead – Product BU (Core Cloud, Multi Cloud & Media Services)	Lead – Product BU (Core Cloud, Multi Cloud & Media Services)